ENGINEERING PORTFOLIO

Ariadne Colina-Valeri

Hi! My name is Ariadne, and I'm a Junior studying Electrical Engineering at Tufts University. I have experience in signal processing, embedded systems and machine learning, from restoring RF signals with a Wiener filter to building a real-time audio effects board and an autonomous robot. I'm looking for internships in those fields.

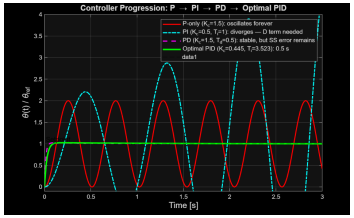


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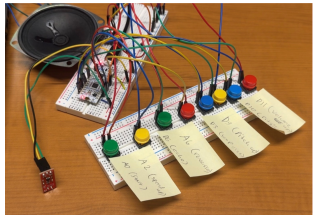
SELECTED PROJECTS



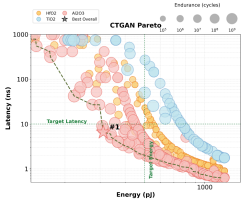
Autonomous Robot
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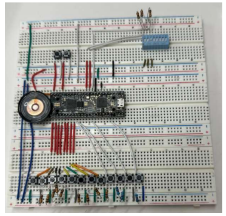
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PROJECT 1 | IN PROGRESS | JUNIOR DESIGN | SEPT. 2026 TO PRESENT

Autonomous ESP32 Robot

THE CHALLENGE

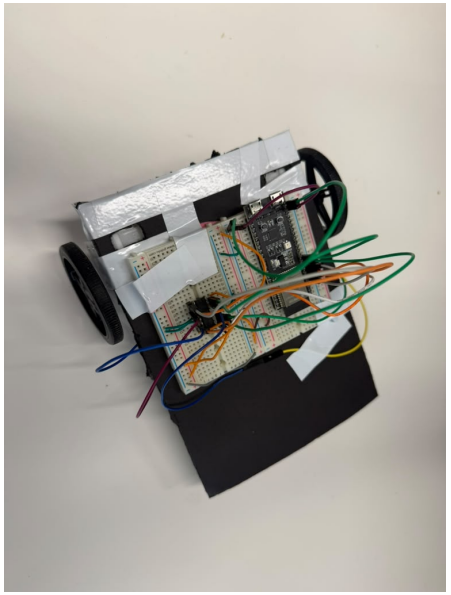
The robot has to follow a colored lane and avoid obstacles with no one controlling it, and accept commands only from our own team.

WHAT I'M DOING

I built a 7-state state machine on the ESP32 that authenticates every websocket command with our team key and rejects the rest. For obstacles, an IR LED and photodetector measure reflected light. Each reading subtracts the room's background light, and the robot stops only when the signal rises 6 standard deviations above its normal noise. Now I'm building the lane color sensing. Colored LEDs light the track, and the robot compares how much of each color reflects back to tell black, yellow, red and blue apart in any room lighting.

NEXT

Assembling the full robot inside the ladybug body I'm designing in SolidWorks, with a custom ESP32 PCB, using parts chosen from datasheets, in place of the breadboard. Then programming lane following, so the robot corrects its steering whenever it drifts off its lane. For the go-beyond part, sound will control its movements, with volume and certain notes each doing something different.



The robot so far, wired on a breadboard around the ESP32



The ladybug body the robot will be assembled in

UAV Pitch Control and RF Signal Restoration

THE CHALLENGE

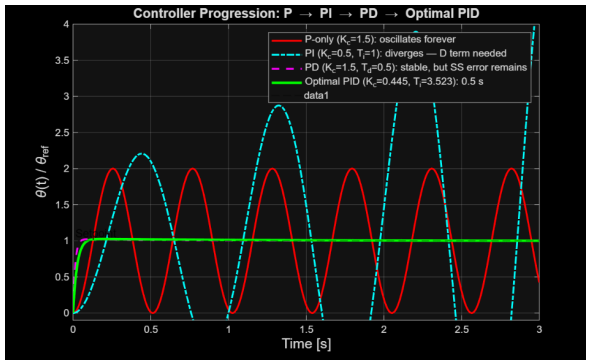
A UAV that shakes while collecting RF signals blurs them, so it has to stay level before the signals can be trusted.

WHAT I DID

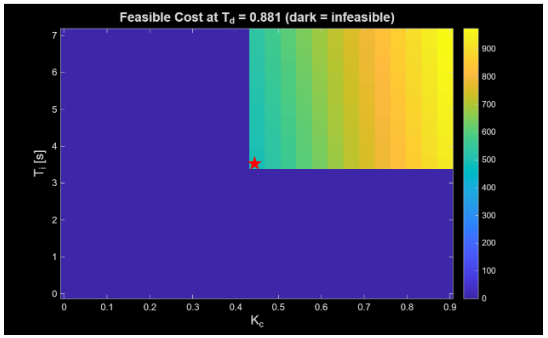
First, I used Routh-Hurwitz analysis to prove that P and PI controllers can never keep the UAV stable, so it needed a full PID controller. I then tested 15,625 combinations of PID settings to find the best one. Next, I restored a blurred RF signal with a Wiener filter in MATLAB. Finally, I showed how the two parts connect, because the longer the controller takes to settle, the longer the signal stays blurred.

THE RESULT

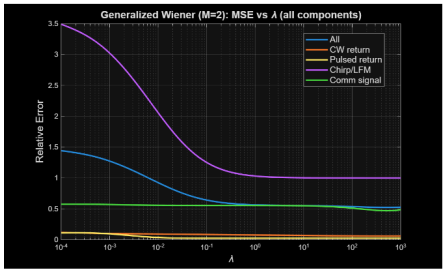
The PID controller rises in 52ms with only 2.6% overshoot and an 88.3° phase margin. The Wiener filter cut the error by 77% compared with a direct inverse, and a controller that settles in 2s instead of 0.5s makes the recovery 1.5× worse.

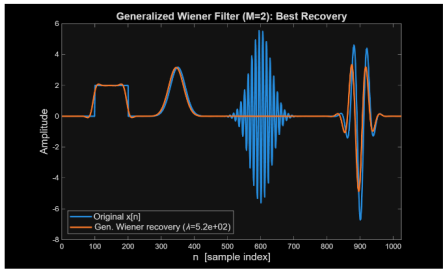


P, PI, PD and PID controllers



Cost of each gain pair; the star is the optimum





Wiener filter error (left) and the restored RF signal over the original (right)

DJ Board on an STM32

(Audio Recorder and Effects Processor)

THE CHALLENGE

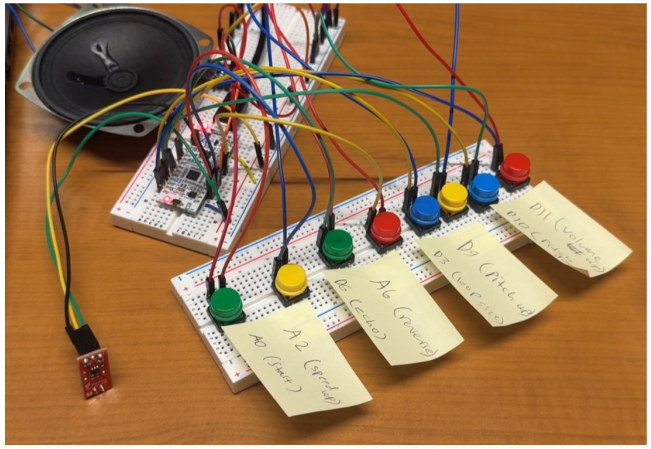
The board has to record sound and play it back in real time with 7 effects (echo, reverse, speed, loop size, pitch up, pitch down, volume), all within the STM32's 64KB of on-chip memory.

WHAT I DID

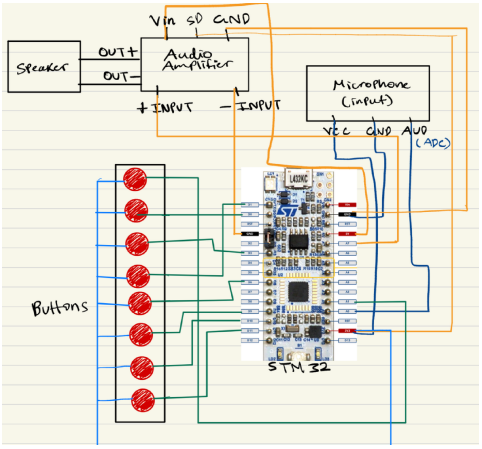
Memory set the sample rate first. At 8kHz only 2 seconds of audio fit, so I recorded at 4kHz to get 5 seconds. Since every effect depends on a clean recording, the next problem to solve was that playback came out with a high-pitched hiss and clicks. To fix the hiss, a 4kHz timer interrupt drives a state machine so every sample is recorded and played at the same rate, and a longer ADC sampling time (12.5 cycles) gives each microphone reading time to settle. The clicks came from sudden jumps in the speaker signal, so I held the output at mid-rail between clips and faded each clip in and out over 256 samples (64ms).

THE RESULT

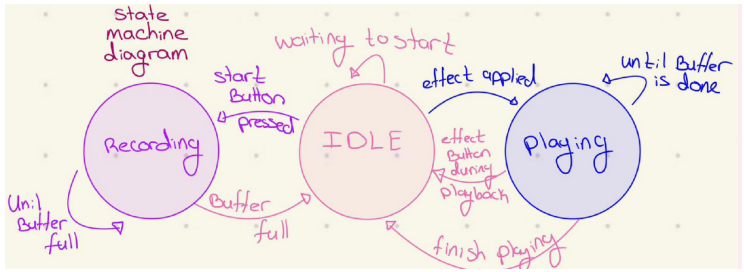
The board records 5 seconds of 12-bit audio in a 40KB buffer, plays all 7 effects, and its playback matched the original recording on the oscilloscope.



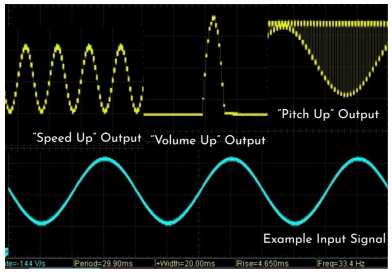
The board



Hardware: wiring diagram



Software: the state machine



Oscilloscope captures of the effects



CTGAN Design Discovery for RRAM

THE CHALLENGE

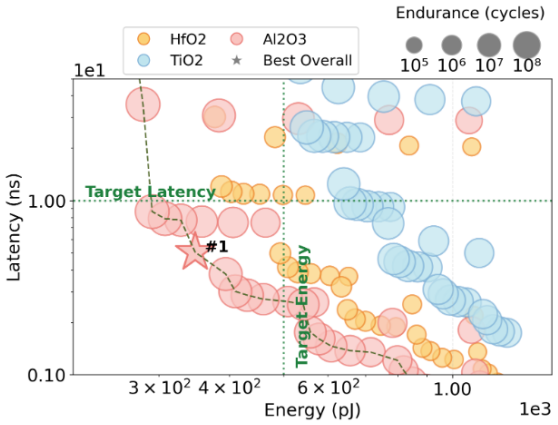
The published framework I built on used a CVAE, an AI model that averages its training data. Averaging hides the unusual designs, and those are where the best performance is.

WHAT I DID

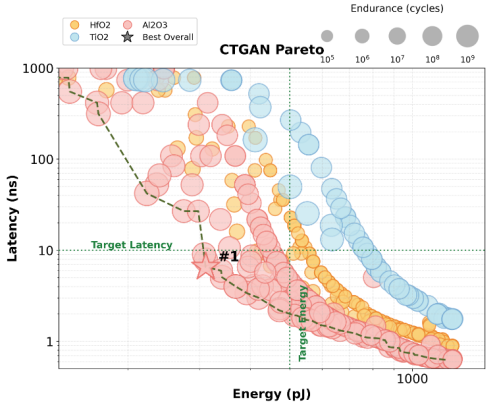
I replaced it with a CTGAN, a model that searches those unusual designs more aggressively. To keep it from inventing designs that cannot physically exist, I used the framework's physics model (PINN) to reject them. The CTGAN failed with 500 training samples and was still noisy with 2,500, so I grew the data to 5,000, where it worked for all 3 materials (Al_2O_3, HfO_2 and TiO_2).

THE RESULT

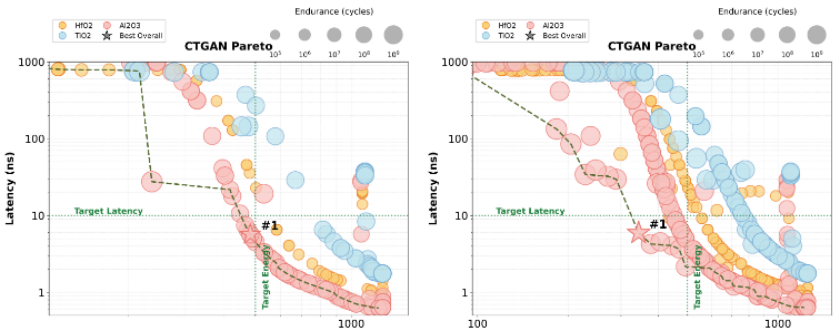
My best Al_2O_3 design lasted 25% longer than the published one (4.84 billion cycles against 3.85 billion) and used 11% less energy, though it switched a little slower (6.57ns against 4.07ns).



Published CVAE results



My CTGAN with 5,000 samples



The CTGAN with 500 samples (left) and 2,500 (right)

PROJECT 5 | DIGITAL LOGIC CIRCUITS | SEPT. 2025 TO DEC. 2025

FPGA Audio Synthesizer

THE CHALLENGE

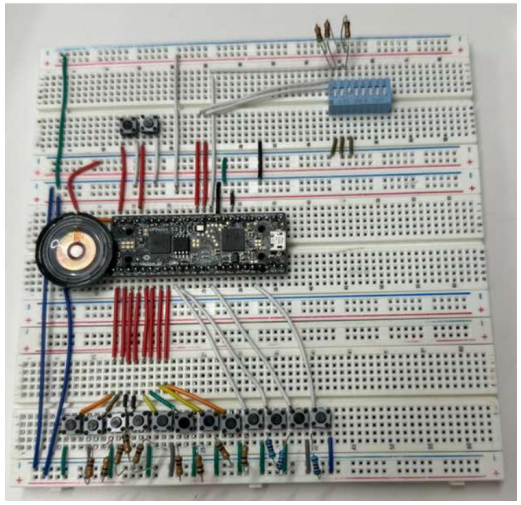
The synthesizer has to play notes and chords from a MIDI keyboard across 8 octaves, using hardware I described in SystemVerilog on an FPGA.

WHAT I DID

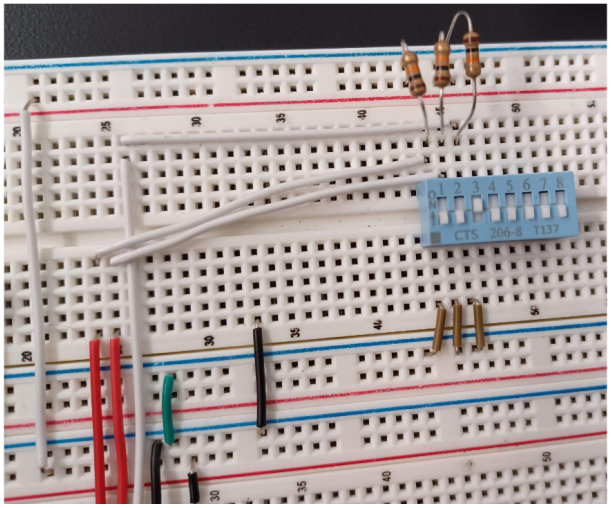
My first design calculated every note by dividing on the chip, and division takes so much hardware that it needed over 250% of the FPGA's space. Each octave doubles the frequency, and dividing by 2 is just a bit shift, so I calculated one octave by hand and got the other seven with bit shifts, which fit the design on the chip. A tuner then showed every note an octave too low, which meant the clock ran at twice the rate the code assumed, so I rebuilt the scale for the real clock. The lowest note was still F#2 (about 92Hz), so I widened the period counter to reach 16Hz. Chords of more than 2 notes, or in low octaves, turned into static, so I plotted chords in Desmos to see the problem, then mixed the notes with PWM on a 3-bit bus, the best balance of volume and clarity.

THE RESULT

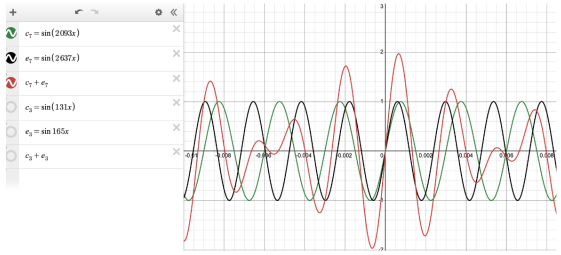
The keyboard plays all 12 notes across 8 octaves (16Hz to 8kHz), chords of 5 to 6 notes in the lowest octave, and every note at once in the highest.



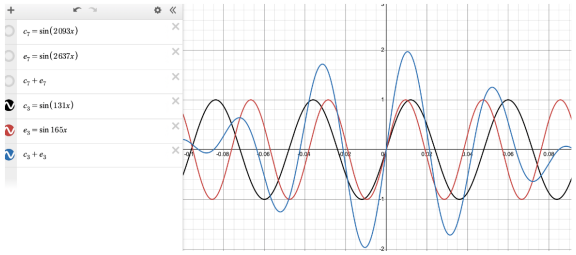
The complete board



The switch that selects the octave



C + E summed in octave 7 (red)



C + E summed in octave 3 (blue)